

Answer D

- 9) useful power output = $mgh / t = 1400 \times 9.81 \times 1.6 = 21974 \text{ W}$
since the motor is 20% efficient, power input = $21974 / 0.2 \text{ W} = 109872 \text{ W}$

Answer C

- 10) Initially, at the top of the jump, $KE = 0$, $EPE = 0$ and GPE is the maximum at 120 kJ